Norwegian University of Science and Technology (NTNU)
Department of Electric Energy (IEL)

TET4565: Electricity Markets and Energy System Planning, Specialisation Course

Course project

In this project you are encouraged to work with a problem formulation of your own choice. You may choose a variant of the problem you are looking at in your project thesis, or another topic you are interested in. The project shall be done in groups of 3 students, and you keep the same group within this project and the project in the other module. All participants of the groups must be able to explain any part of analysis, results and code. The teaching team might call for individual interview if we suspect uneven distribution. The problem you choose to work on should be relevant to energy markets, production planning, flexibility in grid operations or power system expansion. See available list of literature for inspiration. In addition, the problem you choose to work on must fulfil certain requirements:

- Uncertainty must be present.
- The optimisation problem must be linear, either by itself or by approximation.
- The problem should have recourse opportunities, i.e. when you observe the outcome of the uncertain variable, you should be able to do something.
- The problem should preferably only have one state variable, to keep it simple enough.

If you cannot find a suitable problem on your own, you may work on the short term hydro power scheduling problem presented at the end of this note. **Please note:** Keep the problem simple; you will not be graded based on the complexity/coolness of the problem, but how you understand it and use it within this project.

Deadlines:

- Part 1: 18.9.2025 23:59
- Part 2: 02.10.2025 23:59
- Part 3: 23.10.2025 23:59
- Final submission: 27.11.2025 23:59

Part 1: Problem formulation and data cleaning

Define the problem you have chosen for your assignment as a two-stage optimization problem! In addition, you should do a data analysis exercise. If you have a dataset relevant for your problem, work with that, otherwise, choose one of the available datasets. We recommend you to focus on a problem that is understandable and not too complicated. If you plan to make this relevant to your master thesis topic, start simple and make this project the basis for further expansion afterwards within your thesis.

Learning goals:

- Motivate the value of formulating and solving a decision problem, and identifying where how the problem compares with the research frontier.
- Mathematical formulation of a decision problem under uncertainty
- Identify relevant uncertainty and the underlying information structure: When does relevant information becomes available?
- Process, clean, transform, plot and describe relevant data, preferably within the Pandas framework in Python.

Tasks:

- Briefly describe the optimization problem that you are aiming to solve including:
  - Brief context. What background information is needed to understand the problem?
  - Describe the problem intuitively in words.
  - What is the value of solving the problem/understanding the dynamics. Who are the stakeholders?
  - The uncertain variables in the problem, and the associated uncertainty structure. Make sure to also specify the decisions that can be made.
- Summarize the purpose of the model as an explicit research question.
- Write the optimization problem in a mathematical form! Remember to include notation.

Note! Working with a data set relevant for your problem is encouraged. However, as it is early in the semester, you can choose to work with one of the following:

- Your own dataset
- The Rye Microgrid dataset (Dataset A on Blackboard)
- The Austin Household consumption dataset (Dataset B on Blackboard)

The two latter datasets have explicit questions. If you work with your own dataset, make sure to present the following:

- Read in and present the data through relevant plots and/or summarizing statistics
- Look for NaN-values and outliers. Come up with a strategy on how to handle these data. This does not apply if you do not have any outliers or NaN-values.
- Aggregate data, for instance through resampling of date-time-data or grouping.
- Look for pattern in individual data series, and correlation patterns between the series.

Deliverables:

- Formulation of the optimization problem in words, including motivation, value, description of decisions and uncertainty structure. Max 1 page.
- One or a few research questions that you with the help of your model should answer
- Mathematical description of the optimization problem, including notation
- Code for data handling
- Brief description of data. Max 1–2 pages, including plots. Do not answer the questions related to the dataset as numbered bullet points, instead write a summary that includes the answers.

The two last can be combined as a Jupyter notebook, if preferred.

Part 2: Programming a deterministic equivalent

Use programming language to formulate and solve your problem. It is recommended to use the Pyomo package in Python. For a solver, either use the free solver glpk, or obtain an academic license for Gurobi.

If you lack data for your problem, feel free to make up some data. You are allowed to make simple scenarios for your uncertain by defining simple high-low-medium scenarios or drawing from a random variable. If you have many scenarios, feel free to reduce the number of scenarios by any method intuitive to you.

We ask you to solve 3 variants of the problem:

- Solve the problem using the expected value of all scenarios
- Solve the problem for each scenario individually
- Solve the problem as a proper stochastic problem, with nonanticipativity between the first and second stage.

Learning goals:

- Writing working and readable optimization code
- Understanding the difference in nature between a stochastic solution to a problem, and multiple deterministic solution

Tasks:

- Write code for solving the 3 versions of the problem.
- Print out and present the results (decisions, objectives) of the 3 variants.

Deliverables:

- Written document with presentation of results, and discussion of the difference between the solution to the 3 variants. (max 2 pages)
- The code created to solve each problem (can divide scripts for each case)

Part 3: Benders decomposition

In the last assignment, you solved the stochastic problem as a deterministic equivalent. If you have many scenarios, or many decision stages, solving the problem as a deterministic equivalent is intractable. The scenario tree can become quite large and difficult to manage. In this part of the assignment, we are going to decompose the problem along the decision stages, using Benders decomposition.

Learning goals:

- Identify the first stage decision variables and understand how these decisions must be made without knowledge about the outcome of the stochastic variables.
- Using functions to parameterise, call and run optimisation problems.
- Formulating cuts

Tasks:

- Formulate a code that can solve your problem using Benders decomposition, constructing a master and subproblem, and defining upper and lower bounds to measure convergence.
- Find the dual multiplier per scenario from the state variable of the first stage. Use this dual multiplier to formulate a cut. Aggregate into one cut per iteration by averaging.
- Solve the problem until convergence.
- Compare the solution to that of the deterministic equivalent in part 2. Comment on the following: Is the solution similar?

Deliverables:

- Mathematical formulation of the problem using Benders decomposition.
- Code that implements the decomposition described above
- A few paragraphs of text that comment on the solution and compare it to the solution in part 2.

Part 4: Stochastic dynamic programming

Stochastic dynamic programming is mostly used to solve multi-stage optimisation problems. In this assignment we are going to use it to solve a two-stage optimisation problem, in a similar fashion as with Benders decomposition.

An important thing to think about is that some of the structure used on Benders decomposition can be reapplied here. Take note of what exactly is different between the methods, and what parts are reusable. NB: You can represent the future using cuts, or by representing it as a piecewise-linear function. You are free to choose your approach.

Learning goals:

- Understanding the difference between dynamic programming and Benders decomposition with regards to discretizing the state space version using dual multipliers to do it.
- Implement stochastic dynamic programming in code.
- Understand how granularity of the discretisation influence the result.

Tasks:

- Use the same example as in Part 2 and 3. However, instead of using Benders Decomposition, you are to use Stochastic Dynamic Programming (SDP).
- Solve the problem for one stochastic scenario, using 10 discrete points for the state variable in the second stage. Explain your results.
- Solve the problem for 3 discrete points and compare the results with b). How does the discretization influence both time management and accuracy of results?
- Extend the problem to include all stochastic scenarios and solve the problem for 10 discrete points.
- Compare your results with Part 2 and Part 3. In your opinion, which of the decomposition techniques works better? What are the pros and cons of each decomposition technique?

Deliverables:

- Code for solving the problem using stochastic dynamic programming.
- Brief summary of results, and comparison to the other optimisation results.
- A reflection on the solution methods. You have probably solved a small-scale version of your problem. Which method would be most suited to solve a full-scale version of your problem?

Example: Short-term hydropower optimisation problem

This is a problem you can work on if you can't think of another suitable problem for these assignments.

You, as the newest hydropower station employee, are tasked with finding the optimal strategy for the station for the next 48 hours. The task specifically is that you must find the optimal production profile for these 48 hours, based on the estimated inflow and electricity prices that are to follow. The forecasting-team have given you input data, where the first 24 hours are assumed to be perfectly forecasted, but they have attached 5 different scenarios for the remaining 24 hours. You must therefore find the optimal strategy for the first day, so that you have balanced the risk associated with all 5 scenarios.

Table 1: Hydropower plant parameters.

Parameter	**Value**	**Information**
V_0	3 [Mm^3]	The starting reservoir level for the hydropower plant
V_{MAX}	4.5 [Mm^3]	Rated capacity for the reservoir
Q_{MAX}	100 [m^3/s]	Maximum discharge capacity of water from the reservoir to the hydropower unit
P_{MAX}	86.5 [MW]	Maximum production capacity for the hydropower unit
M3s_TO_MM3	3.6/1000 [Mm^3/m^3]	Conversion factor from cubic meter per second to million cubic meter
E_{conv}	0.657 [kWh/m^3]	Power equivalent from discharge water to produced electricity
WV_{end}	52 600 [EUR/Mm^3]	Water value for leftover hydropower at the end of the 48th hour

Table 2: Inflow and price assumptions for the 48-hour horizon.

Parameter	**Value**	**Information**
Inflow (first 24 h)	50 [m^3/s]	The inflow for the first 24 hours, given hourly
Inflow (last 24 h)	$10 \cdot s$ [m^3/s]	The inflow for the last 24 hours, given hourly. This includes uncertainty and a 0-based scenario index. For scenario $s = 0$, inflow is $25 \cdot 0$ each hour of the second day.
$N_{\text{scenarios}}$	5	The number of scenarios for the second stage (last 24 hours)
p_{scenario}	0.2	The probability for each scenario
Price	$50 + t$ [EUR/MWh]	Power price for all 48 hours, given as a linearly increasing function. Example: at hour 13, the price is $50 + 13 = 63$.

Inspirational reading list

You are by no means required to read any of the papers below, but some of them might give you ideas for a problem formulation similar to the one you want to work with. You can find these online by searching for them, for instance at Google Scholar.

- T. Trötscher, M. Korpås, “A framework to determine optimal offshore grid structures for wind power integration and power exchange", Wind Energy 14 (8), 977–992.
- L. A. Roald, D. Pozo, A. Papavasiliou, D. K. Molzahn, J. Kazempour, A. Conejo, “Power systems optimization under uncertainty: A review of methods and applications", Electric Power Systems Research, Volume 214, Part A, 2023.
- H. Farahmand, T. Aigner, G. L. Doorman, M. Korpas, D. Huertas-Hernando, “Balancing Market Integration in the Northern European Continent: A 2030 Case Study", IEEE Transactions on Sustainable Energy, vol. 3, no. 4, pp. 918–930, Oct. 2012.
- A. Hentunen, V. Erkkilä, S. Jenu, “Storage system dimensioning and design tool", D6.1, INVADE H2020 project.
- E. F. Bødal, P. Crespo del Granado, H. Farahmand, M. Korpås, P. Olivella, I. Munné, P. Lloret, “Challenges in distribution grid with high penetration of renewables", D5.1, INVADE H2020 project.
- M. Korpås, A. T. Holen, “Operation planning of hydrogen storage connected to wind power operating in a power market," IEEE Transactions on Energy Conversion, vol. 21, no. 3, pp. 742–749, Sept. 2006.
- T. Burandt, B. Xiong, K. Löffler, P.-Y. Oei, “Decarbonizing China’s energy system – Modeling the transformation of the electricity, transportation, heat, and industrial sectors", Applied Energy, Volume 255, 2019.
- O. Wolfgang, A. Haugstad, B. Mo, A. Gjelsvik, I. Wangensteen, G. Doorman, “Hydro reservoir handling in Norway before and after deregulation", Energy, Volume 34, Issue 10, 2009, Pages 1642–1651.
- J. Kong, H. I. Skjelbred, O. B. Fosso, “An overview on formulations and optimization methods for the unit-based short-term hydro scheduling problem", Electric Power Systems.
- A. Gjelsvik, B. Mo, A. Haugstad, “Long- and Medium-term Operations Planning and Stochastic Modelling in Hydro-dominated Power Systems Based on Stochastic Dual Dynamic Programming", in Handbook of Power Systems I, Springer, 2010.
- J. Matevosyan, M. Olsson, L. Söder, “Hydropower planning coordinated with wind power in areas with congestion problems for trading on the spot and the regulating market", Electric Power Systems Research, Volume 79, Issue 1, 2009, Pages 39–48.

- A. Helseth, B. Mo, A. L. Henden, G. Warland, “Detailed long-term hydro-thermal scheduling for expansion planning in the Nordic power system", IET Generation, Transmission & Distribution, 2018, 12(2), pp. 441–447.
- H. Farahmand, S. Jaehnert, T. Aigner, D. Huertas-Hernando, “Nordic hydropower flexibility and transmission expansion to support integration of North European wind power", Wind Energy 2015; 18(6), pp. 1075–1103.
- K. E. Thorvaldsen, M. Korpås, K. B. Lindberg, H. Farahmand, “A stochastic operational planning model for a zero emission building with emission compensation", Applied Energy, Volume 302, 2021.
- J. Widén, M. Lundh, I. Vassileva, E. Dahlquist, K. Ellegård, M. Wäckelgård, “Constructing load profiles for household electricity and hot water from time-use data – Modelling approach and validation", Energy and Buildings, 41(7), 2009, pp. 753–768.
- I. B. Sperstad, M. Korpås, “Energy Storage Scheduling in Distribution Systems Considering Wind and Photovoltaic Generation Uncertainties", Energies, 2019.
- S.-E. Fleten, T. K. Kristoffersen, “Stochastic programming for optimizing bidding strategies of a Nordic hydropower producer", European Journal of Operational Research, 2007, 181(2), pp. 916–928.
- J. M. Morales, M. Zugno, S. Pineda, P. Pinson, “Electricity market.